

Starting 7:05

Today's Agenda:-

Quizzes On:-

- a) TC & SC
- b) Output Prediction
- c) Debugging Problems

①

$$2n + 100$$

$$T.C \rightarrow O(N)$$

②

$$5n^2 + 3n + 500$$

$$T.C \rightarrow O(N^2)$$

$$\log 1 + \log 2 + \log 3 + \log 4 + \dots + \log N$$

$$\log 1 \times 2 \times 3 \times \dots \times N$$

$$\log N! \quad [1, 2^N]$$

Function mysteryFunction(N):

result = 0

// First loop

For i from 1 to N:

For j from 1 to N:

result = result + 1

$$\} \quad 1_1 \rightarrow N^2$$

// Second loop

For i from 1 to N:

result = result + log(i)

$$\} \quad 1_2 \rightarrow N$$

// Third loop

For i from 1 to 2^N :

result = result + 1

$$\} \quad 1_3 \rightarrow 2^N$$

Return result

$$N = 100$$

$$T.C \rightarrow \mathcal{O}(2^N)$$

$$2^N + N^2 + N$$

$$2^{100} + 10^2 + 100^2$$

①

i	j	Iterations
1	[1, N]	N
2	[1, N]	N
3	[1, N]	N
⋮	⋮	⋮
N	[1, N]	N

$$\left. \begin{matrix} N \\ N \\ N \\ \vdots \\ N \end{matrix} \right\} \underline{N^2}$$

4. **What is the Time complexity?**

```
Function solve(N, M):  
    For i from 1 to N:  
        If N mod i == 0:  
            Print i  
  
    For i from 1 to M:  
        If M mod i == 0:  
            Print i
```

$T.C \rightarrow \underline{O(N+M)}$

5. **What is the Time complexity?**

```
Function func(n):  
    s = 0  
    For i from 1 to 100:  
        s = s + i  
    Return s
```

$[1, 100]$

$\underline{100}$
 $T.C \rightarrow \underline{O(1)}$

What is the Time Complexity of the code?

Function func(n):

s = 0

i = 0

While i < n:

s = s + i

i = i * 2

Return s

5

$s = \phi \phi 0$

$i = \phi \phi 0$

∞

T.C $\rightarrow O(\infty)$

for (i = 0; i ≤ N-1; i++) {
 for (j = 0; j ≤ i; j++) {
 print (i+j);
 }
}

What is the time complexity of the following code?

For i from 0 to N - 1:

For j from 0 to i:

Print(i + j)

i	j	# Iterations
0	[0, 0]	1
1	[0, 1]	2
2	[0, 2]	3
⋮	⋮	⋮
N-1	[0, N-1]	N

1 + 2 + 3 + 4 + ... + N

$$\frac{N(N+1)}{2}$$

$$\frac{N^2}{2}$$

+

$$\frac{N}{2}$$

$$T.C \rightarrow O(\underline{N^2})$$

8.

What is the time complexity of the following code?

For **i** from 1 to 100 step **i * 2**:

For **j** from 1 to **N**:

Print(**i + j**)

for (i = 1 ; i ≤ 100 ; i = i * 2) {

for (j = 1 ; j ≤ N ; j++) {

print()

}

}

<u>i</u>	<u>j</u>	# <u>Iterations</u>
1	$[1, N]$	N
2	$[1, N]$	N
4	$[1, N]$	N
8	$[1, N]$	N
16	$[1, N]$	N
32	$[1, N]$	N
64	$[1, N]$	N
128		

$$7 \times N$$

$$T.C \rightarrow O(N)$$

q.
=

$$f_1 \rightarrow 10 \cdot N$$

$$3 \dots$$

$$f_2 \rightarrow N$$

$$10$$

$$f_3 \rightarrow N \cdot 10 \cdot N$$

$$10 \times 3 \dots$$

$$f_4 \rightarrow N!$$

$$10 \times 9 \times 8 \times 7 \times 6 \times \dots \times 1$$

$$N=10$$

10.

$$f_1 \rightarrow N!$$

$$10 \times 9 \times 8 \times 7 \dots 1$$

$$f_2 \rightarrow 2^N$$

$$2^{10}$$

$$f_3 \rightarrow N^N$$

$$10^{10}$$

$$f_4 \rightarrow \log N$$

$$3 \dots$$

$$100$$

$$f_1 \rightarrow 100 \times 99 \times 98 \times \dots \times 1$$

$$f_2 \rightarrow 2 \times 2 \times 2 \times \dots \times 2$$

$$f_3 \rightarrow 100 \times 100 \times 100 \times \dots \times 100$$

$$f_4 < f_2 < f_1 < f_3$$

11.
Function f1(n):

result = 0

For i from 1 to n:

j = 1

While j < n:

result = result + 1

j = j * 2

Return result

Function f2(n):

result = 1

For i from 1 to n:

result = result * 2

Return result

Function f3(n):

result = 1

For i from 1 to n:

For j from 1 to n:

result = result * n

Return result

Function f4(n):

count = 0

i = 1

While i <= n:

count = count + 1

i = i * 2

Return count

$$f_4 < f_2 < f_1 < f_3$$

for (i = N ; i > 1 ; i = i / 2) {
 |
 print --
 3

100
50
25
12

$$N \xrightarrow{1} \frac{N}{2} \xrightarrow{2} \frac{N}{4} \xrightarrow{3} \frac{N}{8} \dots \xrightarrow{k} 1$$

$$N \rightarrow \frac{N}{2^1} \rightarrow \frac{N}{2^2} \rightarrow \frac{N}{2^3} \xrightarrow{k \text{ iterations}} \frac{N}{2^k}$$

$$\frac{N}{2^k} = 1 ; \therefore N = 2^k$$

Taking log both sides,

$$\log_2 N = \log_2 2^k \\ = k.$$

$$\therefore \boxed{k = \log_2 N}$$

$f_1 \rightarrow i \rightarrow 1$ to N
 $\hat{j} = 1;$
while ($\hat{j} < N$) {
 —
 $\hat{j} = \hat{j} * 2;$
}

i	j	# Situation
1	$C(1, N); \hat{i} = \hat{i} * 2$	$\log N$
2	$C(1, N); \hat{i} = \hat{i} * 2$	$\log N$
$\vdots$	$\vdots$	$\vdots$
N	$C(1, N); \hat{i} = \hat{i} * 2$	$\log N$

$$\underline{\underline{N \cdot \log N.}}$$

12.

What will be the output of the following function?

```
void example() {
    int x = 10;
    for (int i = 1; i <= 5; i++) {
        x -= 2;
        if (x % 3 == 0) {
            continue;
        }
        print(x);
    }
}

example();
```

$x = 10$
~~8~~
~~6~~
~~4~~
~~2~~
 0

$i = 1$
~~2~~
~~3~~
~~4~~
~~5~~
~~6~~

o/p:- 8 4 2

13.

What will be the output of the following function?

```
void processLoop() {
    for (int i = 1; i != 10; i += 2) {
        print(i);
    }
}

processLoop();
```

Run
Indefinitely.

$i = 1$ ~~3~~ ~~5~~ ~~7~~ ~~9~~ ~~11~~ 13
 - - -

o/p:- 1 3 5 7
 9 11 13
 - -

14.

What will be the output of the following function when executed?

```
void example() {  
    for (int i = 1; i <= 4; i++) {  
        if (i % 2 == 0 && i * 2 < 7) {  
            print(i);  
        }  
    }  
}  
example();
```

$i = 1, 2, 3, 4, 5$

o/p:-
2

15.

What is the issue in the following function?

```
void print_till_5() {  
    int i = 1;  
    while (i < 5) {  
        print(i);  
    }  
}  
print_till_5();
```

Run indefinitely.

1 1 1

i is never getting updated.

16

What is the output in the following function?

```
void find_sum() {
    int sum;
    for (int i = 1; i <= 5; i++) {
        sum += i;
    }
    print(sum);
}
find_sum();
```

Sum is not initialised.

17.

What will be the output of the following function?

```
void example2() {
    for (int i = 1; i <= 3; i++) {
        for (int j = 3; j >= 1; j--) {
            print(i + " " + j + " ");
        }
        print("\n")
    }
}
example2();
```

o/p:-

```
1 3 1 2 1 1
2 3 2 2 2 1
3 3 3 2 3 1
```

i

1 2 3

j

3 2 1 0

18

What is the output in the following code?

```
void solve() {  
    int[] arr = {1, 2, 3, 4};  
    for (int i = 0; i <= arr.length; i++) {  
        print(arr[i]);  
    }  
}  
solve();
```

4.

$i = (0, 4)$

arr[4].

0, 1, 2, 3

Array Index Out of Bound.

19.

What issue exists in the following function?

```
void solve() {  
    for (int i = 2; i <= 10; i+=2) {  
        print(i);  
    }  
}  
solve();
```

2, 4, 6, 8, 10, ~~12~~

20.
=

What issue exists in the following function?

```
void solve() {  
    for (int i = 10; i > 0; i++) {  
        print(i);  
        i--;  
    }  
    solve();  
}
```

$i = 10, 11, 12, 13, 14, \dots = -$

21.
=

What issue exists in the following function?

```
void solve() {  
    int[] arr = {5, 10, 15, 20};  
    for (int i = arr.length; i >= 0; i--) {  
        print(arr[i]);  
    }  
    solve();  
}
```

↑
arr out

0, 1, 2, 3

$i = 4$

↑
invalid
index

7 - 9:30

→ Post your code to Discord group.

→ TH request / Scala AI

→ Tag me on Discord & I'll help debugging.

Conceptual Doubt



Marshall Gay

Academ Apr 26 9:00 AM